

Pathway-Level Characterisation of Peripheral Blood Transcriptomic Dysregulation in Spinal Muscular Atrophy

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01 ABSTRACT

Spinal muscular atrophy (SMA) arises from depletion of survival motor neuron (SMN) protein, yet the biological consequences of this deficiency in accessible peripheral tissue remain incompletely characterised at the pathway level. This investigation applied differential expression analysis, followed by Gene Ontology (GO) over-representation analysis (ORA) and complementary gene-set enrichment analysis (GSEA), to whole-blood RNA-seq data from a publicly available dataset on Gene Expression Omnibus (GEO) to determine which cellular compartments are disproportionately represented among the transcriptional differences separating SMA from healthy blood. Differential expression analysis identified 78 upregulated and 22 downregulated candidate genes. GO Cellular Component was the only ontology to yield statistically significant enrichment, producing 13 significant UP-associated terms and 16 significant DOWN-associated terms, while Biological Process and Molecular Function returned no significant categories in either direction. Downregulated genes were dominated by SMN-associated nuclear structures: the SMN complex, SMN-Sm protein complex and Cajal body, along with a parallel muscle-structural signature, both likely driven by SMN1 and SMN2. Upregulated genes were concentrated within extracellular matrix, blood microparticle and membrane/vesicle-associated compartments. Complementary GSEA confirmed transcriptome-wide, same-direction convergence for 6 of these categories. Together, these findings characterise a coherent, pathway-level transcriptomic signature associated with SMA in peripheral blood, which may form a basis for biomarker or therapeutic studies and future replication.

02 RESEARCH GAP & OBJECTIVES

RESEARCH GAP

The consequences of SMN deficiency in SMA are rarely characterised at the level of whole biological pathways, beyond the well-established motor-neuron mechanism

Individual differentially expressed genes are difficult to interpret biologically, while findings are unstable in small cohorts with limited statistical power

Most existing transcriptomic studies have identified isolated candidate genes (e.g. HSPA7) but have not systematically established broader biological disruptions

Reliance on stringent single-gene significance thresholds risks overlooking coordinated but individually modest transcriptional shifts across a shared pathway

RESEARCH OBJECTIVES

1. Identify genes differentially expressed between SMA and healthy whole-blood samples in the G17 cohort using a variance-moderated linear-model framework
2. Utilise 3 categories of GO analysis to determine associated pathways of dysregulated genes, and assess concordance using transcriptome-wide enrichment via GSEA
3. Organise statistically significant pathway-level findings into coherent biological themes and relate them to established SMA biology
4. Identify candidate pathways that may warrant further investigation as biomarkers or therapeutic targets

03 METHOD

Differential Expression by Limma

Whole-blood RNA-seq expression data from the GEO Accession Number: GSE174056 (G17) dataset, which included a cohort of 5 SMA and 5 healthy-control samples, were quality-controlled before statistical testing. Genes absent in more than half of samples were removed, Ensembl identifiers were stripped of version suffixes, and duplicate identifiers were consolidated by retaining the record with the strongest raw differential-expression evidence. After accounting for and performing variance-stabilising data transformations, Differential expression between SMA and healthy groups was then assessed using the Limma framework in RStudio.

Pathway Enrichment by GO

Genes meeting the predefined criteria of $|\log_2 \text{fold change}| \geq 1$ and raw $P < 0.05$ were classified as upregulated or downregulated and mapped from Ensembl to Entrez identifiers. Using clusterProfiler in RStudio, GO over-representation analysis was performed across all three ontologies, Biological Process, Molecular Function and Cellular Component, using the retained G17 gene set as the background universe. Benjamini–Hochberg FDR correction (adjusted $P < 0.05$) was applied throughout. ORA was used as the primary strategy because it directly tests whether genes crossing the differential-expression threshold are concentrated within defined biological categories, providing an interpretable bridge between the gene-level statistics and pathway-level biology.

Validation by GSEA

To determine whether ORA-significant categories truly reflected broader transcriptomic patterns, complementary GSEA was performed on the complete ranked G17 transcriptome, with genes ranked by the Limma-moderated t-statistic and GO Cellular Component retained as the sole framework to allow direct comparison with the ORA results. A term was classified as convergent when it was ORA-significant, GSEA-significant ($FDR < 0.05$) and directionally consistent across both analyses. Convergence therefore represents complementary analytical evidence and thus can be used to identify which ORA findings warranted the strongest biological interpretation.

Procedure Flowchart

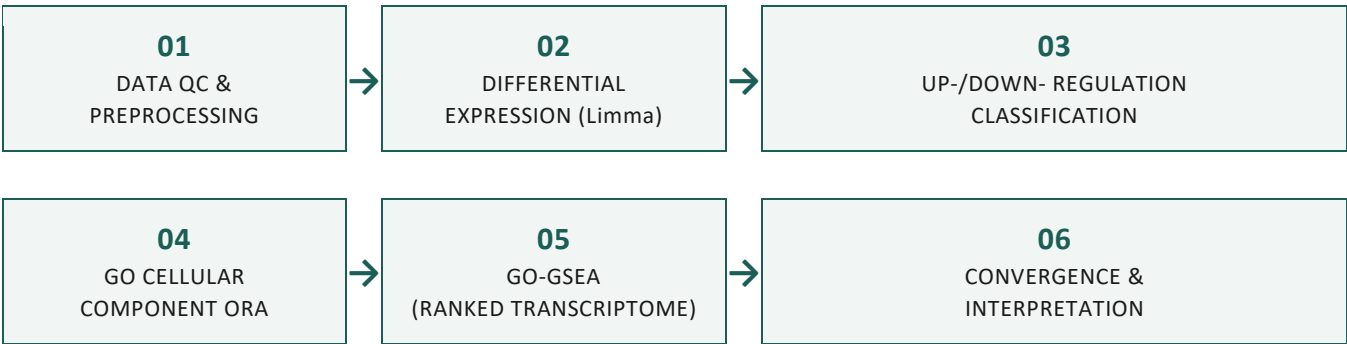


Figure 1. Summary flowchart of the full computational research procedure described above.

04 RESULTS

Differential expression analysis of the G17 whole-blood dataset identified 78 genes meeting the UP criterion and 22 genes meeting the DOWN criterion (Figure 2), of which 63 and 17 respectively mapped successfully to Entrez identifiers for downstream annotation. GO over-representation analysis across all three ontologies produced significant enrichment exclusively within the Cellular Component domain: 13 UP-associated terms (Figure 3), dominated by extracellular matrix, blood microparticle and membrane/vesicle-associated categories, and 16 DOWN-associated terms (Figure 4), dominated by SMN-associated nuclear structures and muscle-structural components. Neither Biological Process nor Molecular Function reached the predefined FDR threshold in either direction, indicating that the candidate gene sets were not sufficiently concentrated within specific molecular functions or processes to survive multiple-testing correction. Complementary GSEA performed on the complete ranked transcriptome confirmed 6 of these Cellular Component categories as convergent (showing the same direction of enrichment at transcriptome-wide scale), comprising of the SMN complex, SMN-Sm protein complex and Z disc among DOWN-associated terms, and extracellular matrix, external encapsulating structure and blood microparticle among UP-associated terms (Figure 5). The analysis outcomes are summarised in Table 1 below.

Table 1. Summary of Differential Expression and Enrichment Outcomes

Metric	Upregulated	Downregulated
Candidate genes ($ \log_2FC \geq 1, P < 0.05$)	78	22
Successfully mapped to Entrez ID	63	17
Significant GO Biological Process terms	0	0
Significant GO Molecular Function terms	0	0
Significant GO Cellular Component terms	13	16
GSEA-convergent Cellular Component terms	3	3

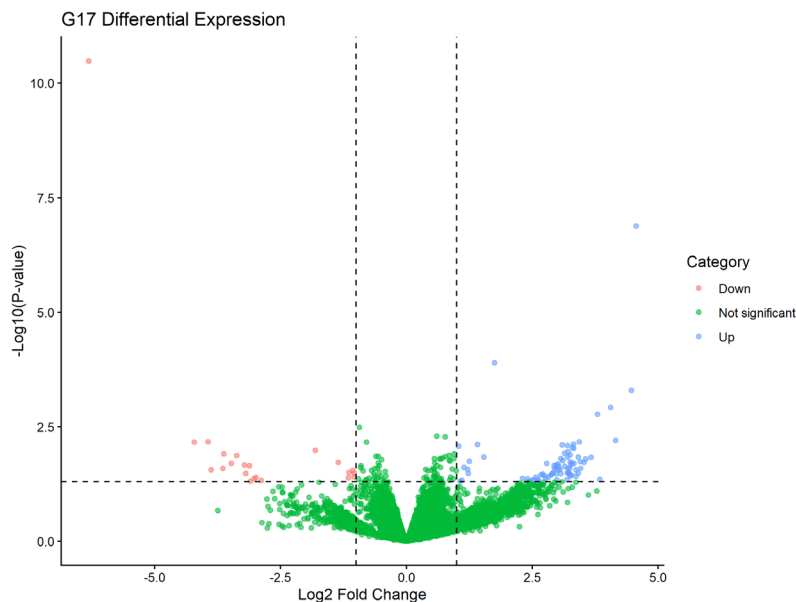


Figure 2. Differential expression of genes in the G17 SMA cohort. Volcano plot of SMA-versus-healthy differential expression following Limma analysis; 78 genes met the UP criterion, and 22 met the DOWN criterion ($|\log_2FC| \geq 1, P < 0.05$).

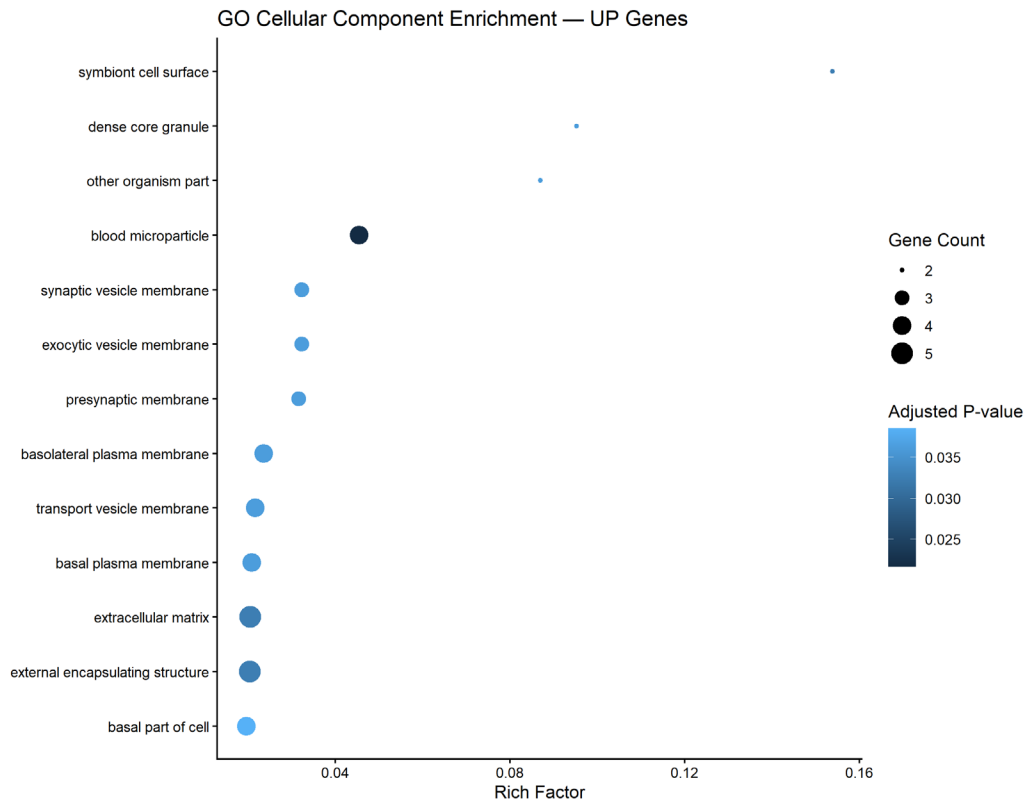


Figure 3. GO Cellular Component enrichment among UP-associated genes: 13 categories reached adjusted $P < 0.05$

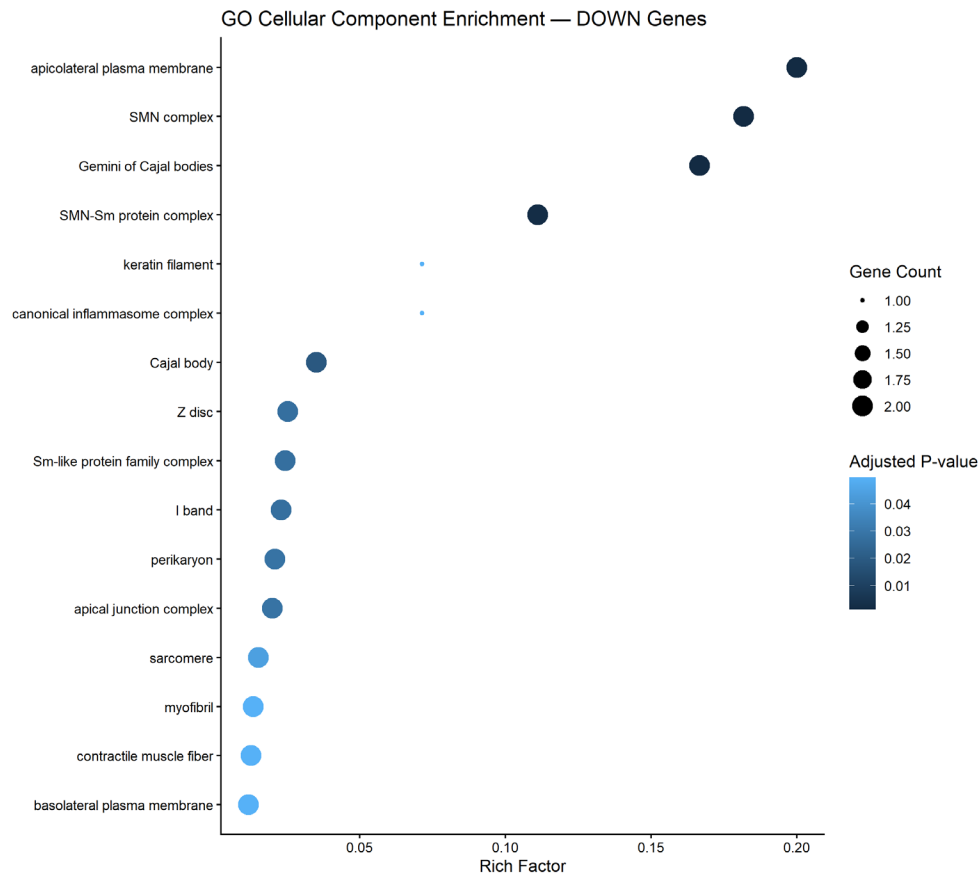


Figure 4. GO Cellular Component enrichment among DOWN-associated genes: 16 categories reached adjusted $P < 0.05$.

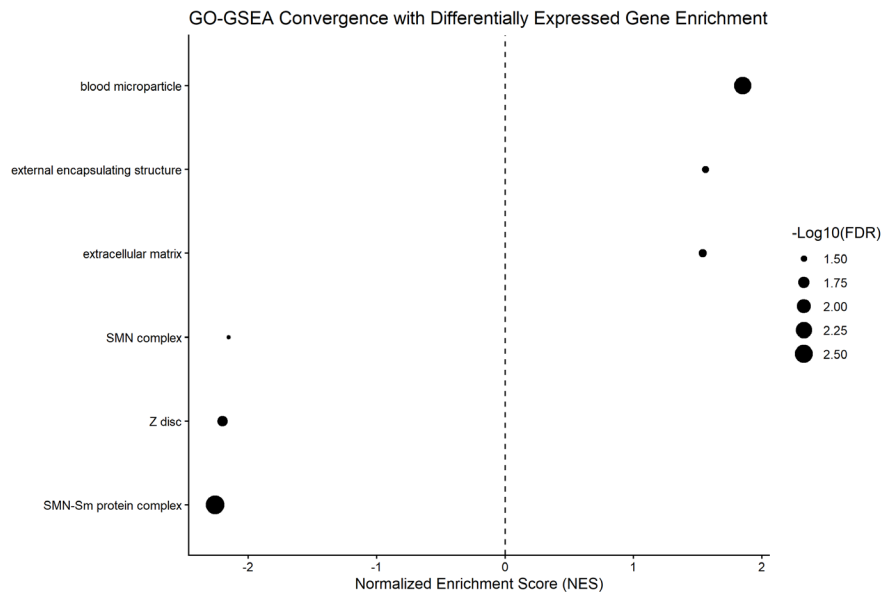


Figure 5. Convergence between GO-ORA and transcriptome-wide GO-GSEA. 6 Cellular Component categories were significant, direction-concordant and shared between both analyses.

05 MAIN FINDINGS

The major transcriptomic findings identified through data analysis are summarised in Figure 6 below.

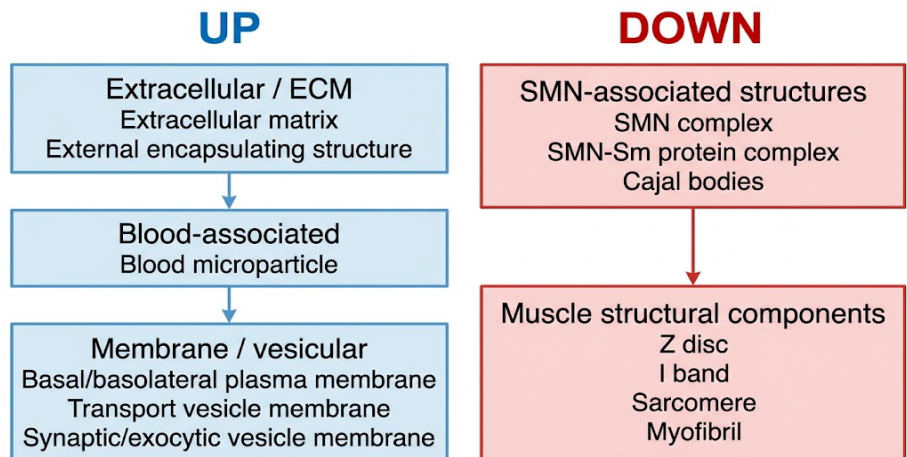


Figure 6. Integrated summary of major transcriptomic signatures identified in G17

Finding 01 — SMN-Associated Nuclear Structures Anchor the Downregulated Signal

The single strongest and most SMA-relevant result in this analysis was the concentration of SMN1 and SMN2 across multiple overlapping DOWN-associated Cellular Component terms, including the SMN complex, SMN-Sm protein complex, Cajal body and Gemini of Cajal bodies. GSEA independently confirmed transcriptome-wide concentration of these categories toward the downregulated end of the ranking (SMN complex: NES = -2.15 , FDR = 0.034 ; SMN-Sm protein complex: NES = -2.26 , FDR = 0.003), which was the strongest convergence observed in the study. Because SMA is fundamentally a disorder of SMN deficiency, this result is not itself novel, but it functions as an internal consistency check: the pipeline correctly recovers the central molecular lesion of SMA from whole-blood RNA-seq, supporting confidence in the study's less-established downstream findings. These

related terms are not independent discoveries, but rather, are best read as a single coherent SMN-related signal rather than separate pathway findings.

Finding 02 — A Convergent Extracellular and Blood-Associated Upregulation Signature

Upregulated candidate genes (C1QB, C1S, ANG, ADAMTSL5, OLFML2A, ITIH4, HSPA7) were significantly concentrated within extracellular matrix, external encapsulating structure and blood microparticle categories, and all three showed statistically significant, same-direction GSEA enrichment across the complete transcriptome (blood microparticle: NES = +1.85, FDR = 0.004; external encapsulating structure: NES = +1.56, FDR = 0.030; extracellular matrix: NES = +1.54, FDR = 0.028). Because these categories reached significance under both threshold-dependent ORA and transcriptome-wide GSEA, this UP-associated extracellular/blood signature is the most statistically robust finding of the study. Elevated circulating extracellular-vesicle and microparticle concentrations have previously been reported in SMA blood and model systems and proposed as pharmacodynamic biomarkers that track with treatment response, lending plausible biological context to this observation. However, GO Cellular Component annotation describes where a gene product is located rather than whether extracellular-matrix or microparticle activity is functionally increased, so the finding should be read as a localisation signature rather than direct evidence of a specific mechanism.

Finding 03 — A Muscle-Structural Signal With One Statistically Convergent Anchor

Downregulated genes were also significantly enriched across a hierarchy of muscle-structural categories — Z disc, I band, sarcomere, myofibril and contractile muscle fibre — all driven by the same SMN1/SMN2 annotation rather than by independent muscle-specific genes. Of these, only the Z disc reached the predefined GSEA significance threshold (NES = -2.20, FDR = 0.021); the remaining related terms showed concordant negative NES values but did not clear FDR < 0.05 (I band FDR = 0.051; sarcomere FDR = 0.074; myofibril FDR = 0.074; contractile muscle fibre FDR = 0.078). This pattern is biologically plausible: SMN protein localises to the sarcomeric Z-disc, and SMA muscle histology shows Z-disc and myofibrillar disorganisation. Yet, the whole-blood tissue source means these annotations reflect a systemic transcriptional footprint of SMN deficiency rather than a direct measurement of muscle tissue.

06 LIMITATIONS OF STUDY

The G17 dataset cohort comprised only five SMA and five healthy-control samples, limiting statistical power and making individual genes and small GO categories disproportionately influential on the enrichment statistics; several significant terms, including the SMN- and muscle-associated categories, were supported by only one or two genes and should not be read as evidence of correspondingly large biological effects. Whole blood does not directly sample motor-neuron or muscle tissue, so the muscle-structural and SMN-associated findings describe a systemic transcriptomic footprint rather than a direct tissue-level measurement. Differential expression demonstrates association rather than causation, and GO Cellular Component annotation describes cellular localisation rather than confirmed pathway activity. Several related GO categories are highly overlapping and driven by the same underlying genes, so the number of significant terms should not be read as the number of independent biological discoveries. The findings were generated from a single dataset and require independent replication before being considered generalisable.

07 AI INTEGRATION

The table below documents which named AI agents contributed to each research stage of this investigation, and their relative share of AI-assisted work within that stage. Bars reflect the proportion of AI usage only and do not track human involvement and oversight.

01 — PROJECT DESIGN



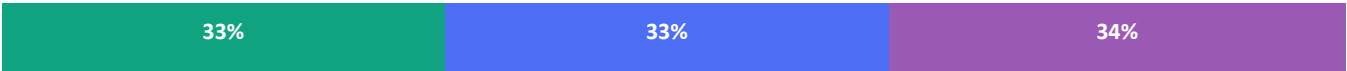
● ChatGPT

02 — RESEARCH PROCEDURE (R STUDIO IMPLEMENTATION)



● ChatGPT

03 — LITERATURE REVIEW & FINDINGS SYNTHESIS



● ChatGPT ● DeepSeek ● Anara

04 — FIGURE GENERATION



● Gemini

05 — REPORT GENERATION



● Claude

08 CONCLUSION

This investigation applied differential expression analysis and complementary GO over-representation and gene-set enrichment analysis to whole-blood RNA-seq data from ten G17 participants (five SMA, five healthy controls) to characterise pathway-level transcriptomic dysregulation associated with SMA. Cellular Component was the only GO domain to yield significant enrichment, identifying a coherent DOWN-associated SMN- and muscle-structural signal alongside a convergent UP-associated extracellular and blood-microparticle signal. GSEA independently confirmed six of these categories as transcriptome-wide convergent, providing complementary evidence that the over-representation findings are not solely an artefact of the chosen differential-expression threshold, and that the pipeline correctly recovers the central molecular lesion of SMA — SMN deficiency — from an accessible peripheral tissue.

Future directions include: independent replication in a larger, clinically annotated SMA cohort, which could assist in determining whether the SMN-associated, muscle-structural and extracellular/blood-microparticle signatures identified here are reproducible features of SMA rather than characteristics of this specific sample. Protein-level and functional validation, together with tissue-specific transcriptomic data, would help establish whether the observed associations reflect functionally meaningful pathway activity, consistent with the portfolio's model of cross-project research development.

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